

Международная олимпиада в сфере информационных технологий «IT-Планета 2025» Конкурс «Современные сетевые технологии» Первый отборочный этап. Задание 2

IT PLANET



Уч? Геворгян Арег

Astra Linux Common Edition

ОС Astra Linux Common Edition, доступная для физических лиц, на данный момент неактуальна, не получает обновления и не лицензируется для использования юридическими лицами.

Чтобы получить актуальную версию ОС Astra Linux Special Edition, направьте нам, пожалуйста, [запрос на тестирование](#) или обратитесь к любому из [официальных партнеров «Группы Астра»](#).

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Create Virtual Machine



Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.

Machine Name and OS Type

Machine Name	server
Machine Folder	E:/VMS/server
ISO Image	E:/alice-2.12.46.6-17.04.2023_15.09.iso
Guest OS Type	Ubuntu (64-bit)
Skip Unattended Install	false

Hardware

Base Memory	4098
Processor(s)	4
EFI Enable	false

Disk

Disk Size	25.00 GB
Pre-allocate Full Size	false



Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are ha



Machine Name and OS Type

Machine Name	client1
Machine Folder	E:/VMS/client1
ISO Image	E:/alce-2.12.46.6-17.04.2023_15.09.iso
Guest OS Type	Ubuntu (64-bit)
Skip Unattended Install	false



Hardware

Base Memory	6033
Processor(s)	4
EFI Enable	false



Disk

Disk Size	20.63 GB
Pre-allocate Full Size	false



Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.



Machine Name and OS Type

Machine Name	client2
Machine Folder	E:/VMS/client2
ISO Image	E:/alce-2.12.46.6-17.04.2023_15.09.iso
Guest OS Type	Ubuntu (64-bit)
Skip Unattended Install	false



Hardware

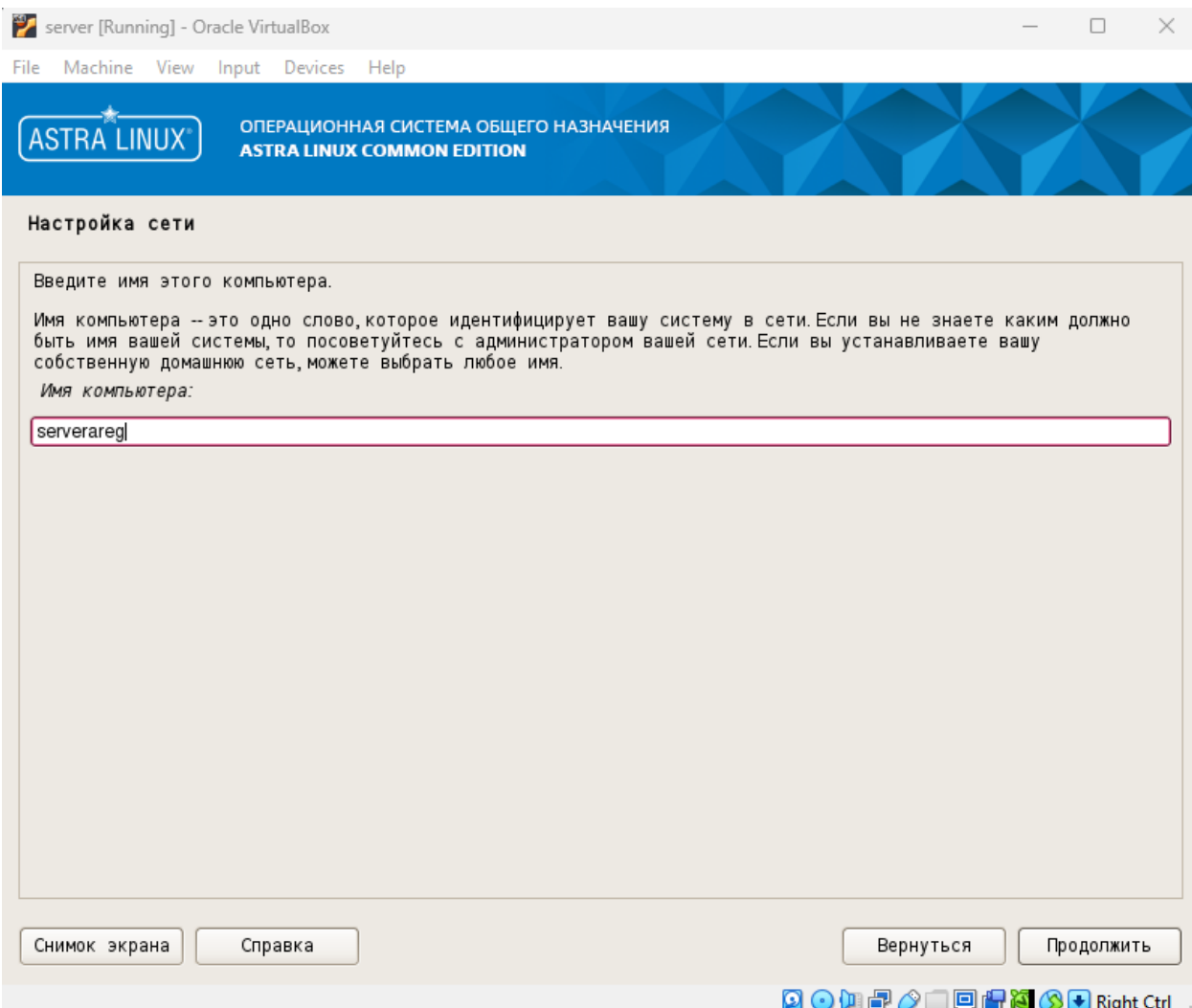
Base Memory	6144
Processor(s)	4
EFI Enable	false

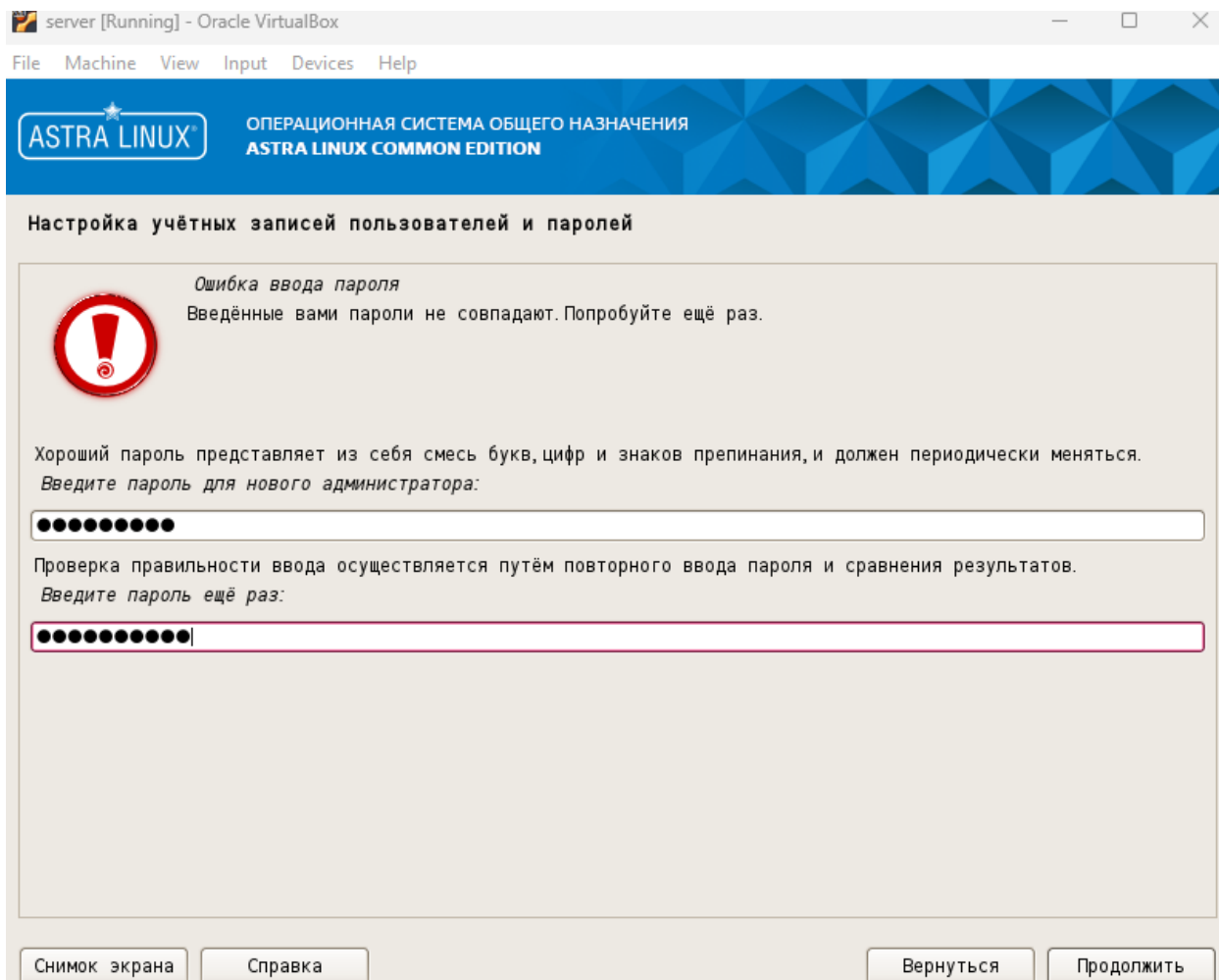


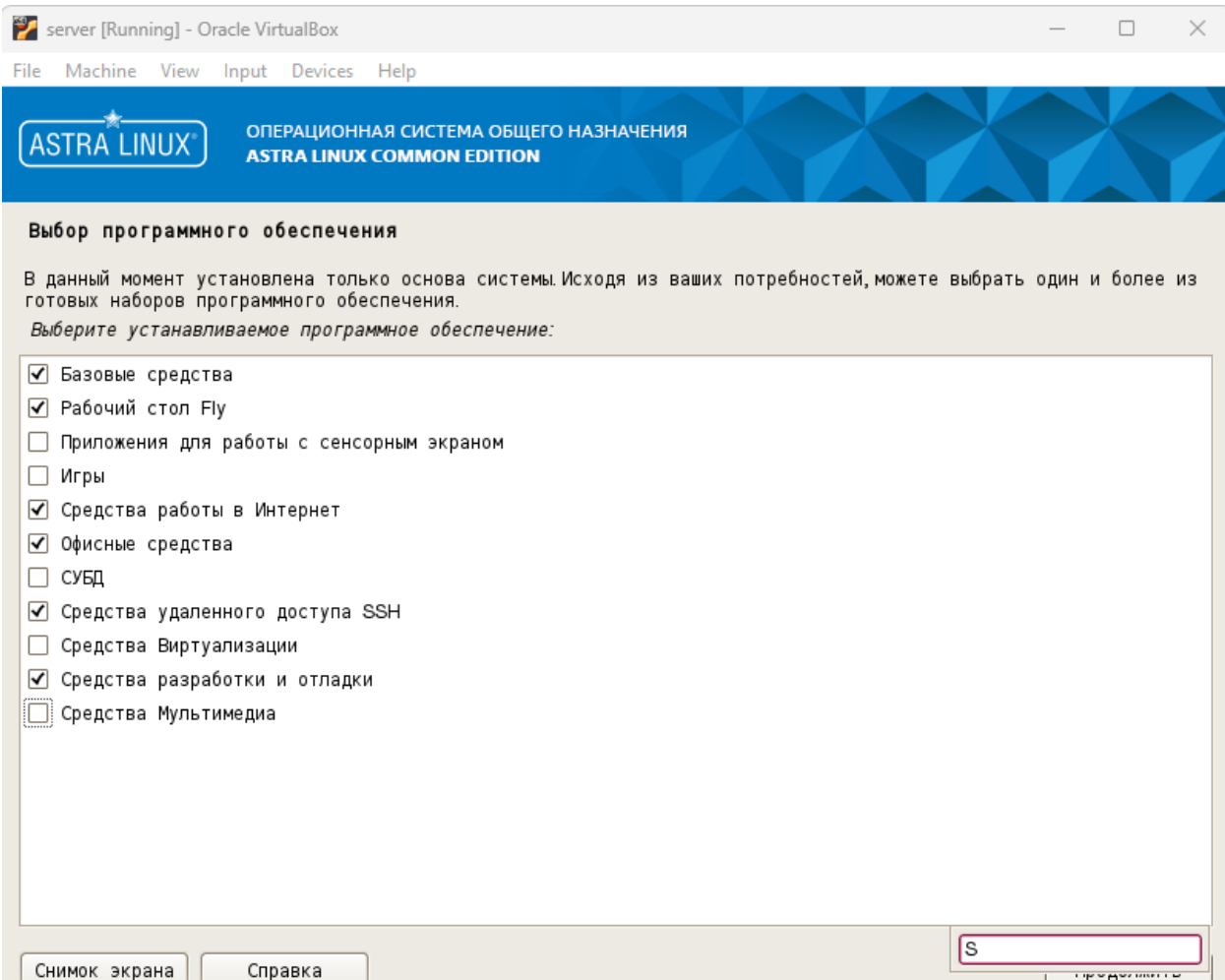
Disk

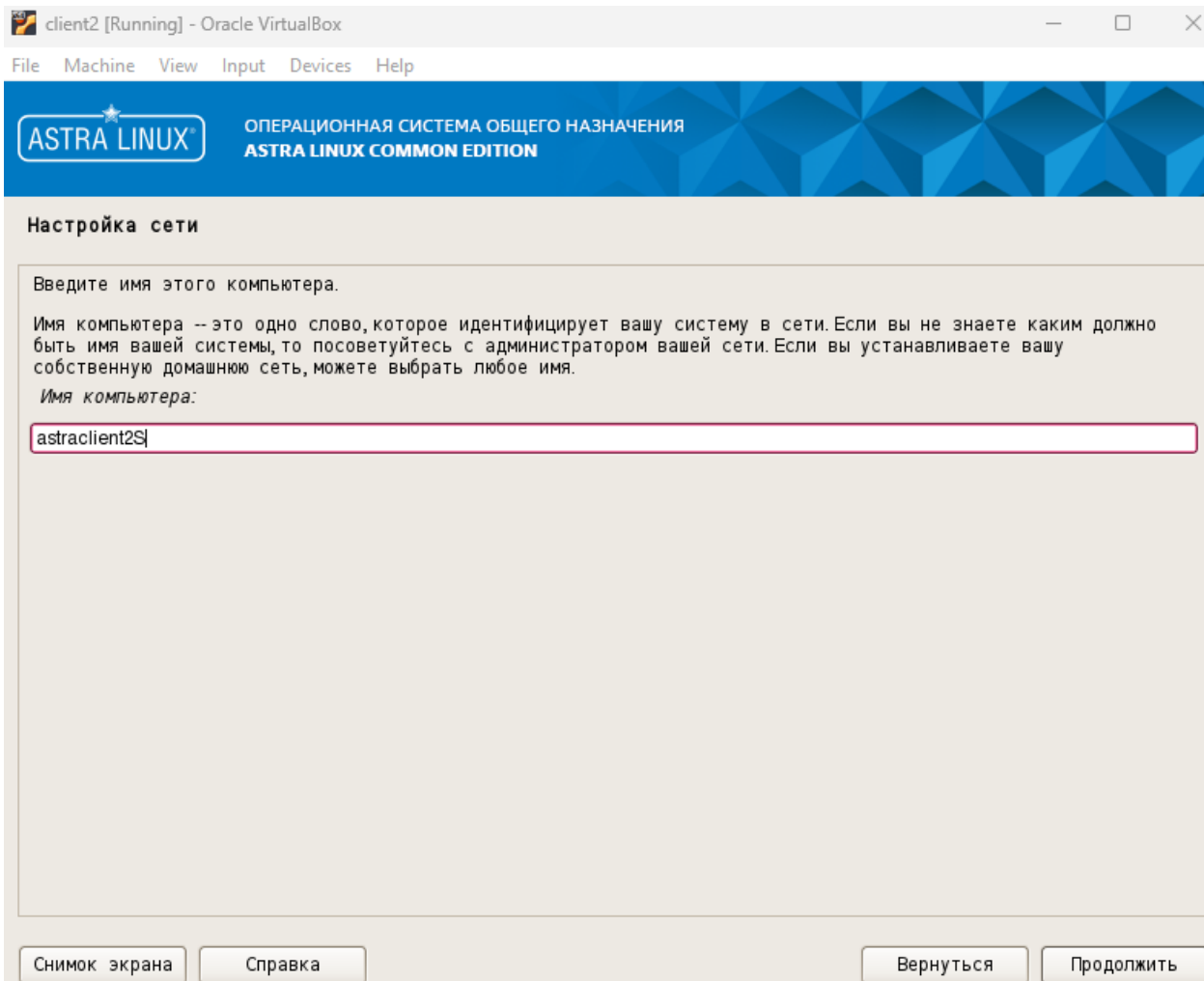
Disk Size	25.00 GB
Pre-allocate Full Size	false

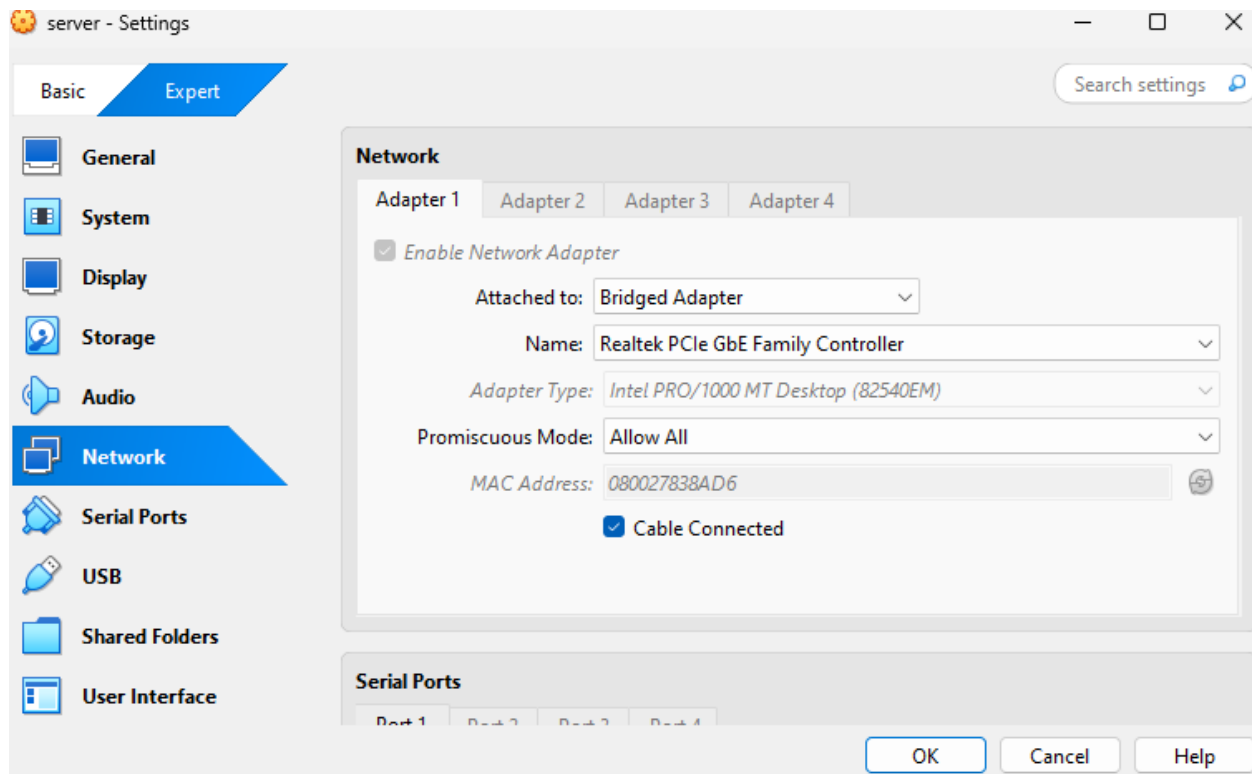
[Help](#)[Back](#)[Finish](#)[Cancel](#)











Установка ансибля у меня он уже был установлен

```
KS000: command not found
on bash
15:23:10 | 2 Apr, Wednesday | in
> sudo apt install ansible
[sudo] password for devkan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (9.2.0+dfsg-0ubuntu5).
The following packages were automatically installed and are no longer required:
  alsa-topology-conf alsa-ucm-conf fish-common libasound2-data libasound2t64 libgcc-14-dev
  libgccjit0 libpcre2-32-0 libtree-sitter0 xsel
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 118 not upgraded.
```

Подключение cxx к серверу

ssh: could not resolve hostname: Name or service not known

on bash

15:08:55 | 2 Apr, Wednesday | in

ssh areg@192.168.5.4

The authenticity of host '192.168.5.4 (192.168.5.4)' can't be established.

ED25519 key fingerprint is SHA256:6t6MP0JAq9CT3kq3PEp36FuueVqZWL35561XMgeaVY

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? y

Please type 'yes', 'no' or the fingerprint: yes

Warning: Permanently added '192.168.5.4' (ED25519) to the list of known hosts

areg@192.168.5.4's password:

Permission denied, please try again.

areg@192.168.5.4's password:

You have new mail.

Last login: Wed Apr 2 13:48:34 2025

areg@serverareg:~\$

что мне дальше на сервере нужно настроить перед все матим

GNU nano 7.2

inventory.ini

[server]

192.168.5.4 ansible_user=areg ansible_ssh_pass=adminareg ansible_become_pass=False

[clients]

192.168.5.5 ansible_user=client1admin ansible_ssh_pass=adminareg ansible_become_pass=False

192.168.5.6 ansible_user=client2areg ansible_ssh_pass=aregadmin ansible_become_pass=False

group

```

core/2.16/reference_appendices/interpreter_discovery.html for more information.
192.168.5.4 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python2.7"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Platform linux on host 192.168.5.5 is using the discovered Python interpreter at
/usr/bin/python3, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
192.168.5.5 | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "msg": "ansible-core requires a minimum of Python2 version 2.7 or Python3 version 3.6. (
current version: 3.5.3 (default, Sep 13 2022, 14:31:32) [GCC 6.3.0 20170516]"
}
[WARNING]: Platform linux on host 192.168.5.6 is using the discovered Python interpreter at
/usr/bin/python3, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
192.168.5.6 | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "msg": "ansible-core requires a minimum of Python2 version 2.7 or Python3 version 3.6. (
current version: 3.5.3 (default, Sep 13 2022, 14:31:32) [GCC 6.3.0 20170516]"
}

```

Решение проблемы с пайтоном

```

areg@serverareg:~$ history
1  python3 --version
2  python --version
3  sudo apt update
4  sudo apt install python3.7 python3.7-venv python3.7-dev
5  python --version
6  sudo update-alternatives --install /usr/bin/python3 python3 /usr/bin/python3.5 1
7  sudo update-alternatives --install /usr/bin/python3 python3 /usr/bin/python3.7 2
8  python --version
9  sudo update-alternatives --config python3
10 python --version
11 python3 --version

```

```

16:09:01 | 2 Apr, Wednesday | in → xsoft_infra
→ ansible all -m ping -i ~/xsoft_infra/inventory.ini
[WARNING]: Platform linux on host 192.168.5.4 is using the discovered Python interpreter at
/usr/bin/python2.7, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
192.168.5.4 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python2.7"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Platform linux on host 192.168.5.6 is using the discovered Python interpreter at
/usr/bin/python2.7, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
192.168.5.6 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python2.7"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Platform linux on host 192.168.5.5 is using the discovered Python interpreter at
/usr/bin/python2.7, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
192.168.5.5 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python2.7"
  },
  "changed": false,
  "ping": "pong"
}

```

на астра линуксе по дефолту стоит 2.7 версия пайтон для ансибля нужна версия постарше
 взялся обновлять все 3 вма поочередно так же все делла через ssh такой проблемы на ОС
 по дфолтку будет стоять пайтон выше версии по дефолту возникать не будет так тчо
 автоматизировать этот процесс я не принялся.

```

areg@serverareg:~$ which python3
/usr/bin/python3
areg@serverareg:~$ python3 --version
Python 3.7.3
areg@serverareg:~$

```

До обновления была версия 2.7

```

client2areg@astraclient2S:~/Загрузки/Python-3.8.10$ python --version
Python 2.7.13
client2areg@astraclient2S:~/Загрузки/Python-3.8.10$ sudo update-alternatives --install /usr/bin/python python
n /usr/bin/python3.7 1
client2areg@astraclient2S:~/Загрузки/Python-3.8.10$ python --version
Python 3.7.3

```

Добавил проверку на версию пайтона

```
GNU nano 7.2 inventory.ini
[server]
192.168.5.4 ansible_user=areg ansible_ssh_pass=adminareg ansible_become_pass=adminareg ansible_python_interpreter=/usr/bin/python3.7

[clients]
192.168.5.5 ansible_user=client1admin ansible_ssh_pass=adminpass ansible_become_pass=adminpass ansible_python_interpreter=/usr/bin/python3.7
192.168.5.6 ansible_user=client2areg ansible_ssh_pass=adminpass ansible_become_pass=adminpass ansible_python_interpreter=/usr/bin/python3.7

[all:vars]
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

ansible-playbook -i inventory.ini @1_server.yml
[WARNING]: Could not match supplied host pattern, ignoring: servers

PLAY [Настройка сервера и сервисов] *****
skipping: no hosts matched

PLAY RECAP *****
```

Исправил ошибку в файле инвентори вместо сервер надо было написать серверс потому что в плейбке крч так хето кдзец сах

```
18:13:14 | 2 Apr, Wednesday | in → xsoft_infra
ansible-playbook -i inventory.ini @1_server.yml

PLAY [Настройка сервера и сервисов] *****

TASK [Гathering Facts] *****
ok: [192.168.5.4]

TASK [Установка необходимых пакетов] *****
changed: [192.168.5.4]

TASK [Установка и настройка Samba] *****
changed: [192.168.5.4]

TASK [Настройка DNS-сервера (Bind9)] *****
changed: [192.168.5.4]

TASK [Настройка DHCP-сервера] *****
changed: [192.168.5.4]

TASK [Настройка веб-сервера Apache] *****
changed: [192.168.5.4]

TASK [Настройка межсетевого экрана (iptables)] *****
changed: [192.168.5.4]

RUNNING HANDLER [restart samba] *****
changed: [192.168.5.4]

RUNNING HANDLER [restart apache] *****
changed: [192.168.5.4]

PLAY RECAP *****
192.168.5.4 : ok=9 changed=8 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

on bash
18:13:52 | 2 Apr, Wednesday | in → xsoft_infra
```

Проверим работу сервисов на сервере

```
areg@serverareg:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2025-04-02 17:13:50 MSK; 3min 36s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 4049 ExecStop=/usr/sbin/apachectl stop (code=exited, status=0/SUCCESS)
   Process: 4056 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 4060 (apache2)
     Tasks: 55 (limit: 4915)
   CGroup: /system.slice/apache2.service
           └─4060 /usr/sbin/apache2 -k start
             └─4061 /usr/sbin/apache2 -k start
               └─4062 /usr/sbin/apache2 -k start

anp 02 17:19:19 serverareg systemd[1]: Starting The Apache HTTP Server...
anp 02 17:19:19 serverareg apachectl[4150]: AH00528: apache2: could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
anp 02 17:19:19 serverareg systemd[1]: Started The Apache HTTP Server.
```

Исправил ошибку где Apache не мог определить полное доменное имя сервера добавив в файл конфигурации ServerName localhost

```
areg@serverareg:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: en
   Active: active (running) since Wed 2025-04-02 17:20:29 MSK; 1s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 4230 ExecStop=/usr/sbin/apachectl stop (code=exited, status=0/SUCCESS)
   Process: 4238 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 4242 (apache2)
     Tasks: 55 (limit: 4915)
   CGroup: /system.slice/apache2.service
           └─4242 /usr/sbin/apache2 -k start
             └─4243 /usr/sbin/apache2 -k start
               └─4244 /usr/sbin/apache2 -k start

anp 02 17:20:29 serverareg systemd[1]: Starting The Apache HTTP Server...
anp 02 17:20:29 serverareg systemd[1]: Started The Apache HTTP Server.
```

Bind9

```
areg@serverareg:~$ sudo systemctl status bind9
● bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2025-04-02 17:13:42 MSK; 11min ago
     Docs: man:named(8)
   Main PID: 3456 (named)
     Tasks: 7 (limit: 4915)
   CGroup: /system.slice/bind9.service
           └─3456 /usr/sbin/named -f -u bind

anp 02 17:13:42 serverareg named[3456]: zone 127.in-addr.arpa/IN: loaded serial 1
anp 02 17:13:42 serverareg named[3456]: zone 255.in-addr.arpa/IN: loaded serial 1
anp 02 17:13:42 serverareg named[3456]: zone localhost/IN: loaded serial 2
anp 02 17:13:42 serverareg named[3456]: all zones loaded
anp 02 17:13:42 serverareg named[3456]: running
anp 02 17:13:51 serverareg named[3456]: validating ./NS: got insecure response; parent indicates it should be secure
anp 02 17:13:51 serverareg named[3456]: insecurity proof failed resolving './NS/IN': 192.112.36.4#53
anp 02 17:13:51 serverareg named[3456]: success resolving './DNSKEY' (in './?') after disabling EDNS
anp 02 17:13:51 serverareg named[3456]: managed-keys-zone: No DNSKEY RRSIGs found for '.': success
anp 02 17:13:52 serverareg named[3456]: resolver priming query complete
```

Samba

```

anreg@serverareg:~$ sudo systemctl status smb
● smb.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smb.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2025-04-02 17:13:50 MSK; 11min ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
 Process: 4018 ExecStartPre=/usr/share/samba/update-apparmor-samba-profile (code=exited, status=0/SUCCESS)
 Main PID: 4021 (smbd)
   Status: "smbd: ready to serve connections..."
    Tasks: 4 (limit: 4915)
   CGroup: /system.slice/smb.service
           └─4021 /usr/sbin/smbd --foreground --no-process-group
             └─4023 /usr/sbin/smbd --foreground --no-process-group
               └─4024 /usr/sbin/smbd --foreground --no-process-group
                 └─4025 /usr/sbin/smbd --foreground --no-process-group

anp 02 17:13:50 serverareg systemd[1]: Starting Samba SMB Daemon...
anp 02 17:13:50 serverareg smbd[4021]: [2025/04/02 17:13:50.070246, 0] ../lib/util/become_daemon.c:138(daemon_ready)
anp 02 17:13:50 serverareg systemd[1]: Started Samba SMB Daemon.
anp 02 17:13:50 serverareg smbd[4021]: daemon_ready: STATUS=daemon 'smbd' finished starting up and ready to serve connections

```

```
PLAY [Настройка сервера и сервисов] *****
TASK [Gathering Facts] *****
ok: [192.168.5.4]

TASK [Установка необходимых пакетов] *****
ok: [192.168.5.4]

TASK [Установка и настройка Samba] *****
ok: [192.168.5.4]

TASK [Настройка DNS-сервера (Bind9)] *****
ok: [192.168.5.4]

TASK [Настройка DHCP-сервера] *****
ok: [192.168.5.4]

TASK [Настройка веб-сервера Apache] *****
ok: [192.168.5.4]

TASK [Настройка межсетевого экрана (iptables)] *****
ok: [192.168.5.4]

PLAY RECAP *****
192.168.5.4      : ok=7   changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

on bash
18:38:07 | 2 Apr, Wednesday | in → xsoft_infra
```

```

- name: Настройка сервера и сервисов
  hosts: servers
  become: true
  tasks:
    - name: Установка необходимых пакетов
      apt:
        name:
          - samba
          - bind9
          - isc-dhcp-server
          - apache2
          - iptables
        state: present
        update_cache: yes

    - name: Настройка контроллера домена (Samba)
      block:
        - name: Установка и настройка Samba
          template:
            src: samba_config.j2
            dest: /etc/samba/smb.conf
          notify:
            - restart samba

    - name: Настройка DNS-сервера (Bind9)
      template:
        src: named.conf.j2
        dest: /etc/bind/named.conf

    - name: Настройка DHCP-сервера
      template:
        src: dhcpd.conf.j2
        dest: /etc/dhcp/dhcpd.conf

    - name: Настройка веб-сервера Apache
      template:
        src: apache_config.j2
        dest: /etc/apache2/sites-available/000-default.conf
      notify:
        - restart apache

    - name: Настройка межсетевого экрана (iptables)
      iptables:
        chain: INPUT
        source: 0.0.0.0/0
        destination: 192.168.5.4
        protocol: tcp
        destination_port: 80
        jump: ACCEPT

    handlers:
      - name: restart samba
        service:
          name: smbd
          state: restarted

      - name: restart apache
        service:
          name: apache2
          state: restarted

    - name: Создание домена XSOFT.LOCAL
      command: samba-tool domain provision --use-rfc2307 --realm=XSOFT.LOCAL --domain=XSOFT --adminpass=Secret123!

```

```

- name: Настройка DNS-сервера (Bind9)
  template:
    src: named.conf.j2
    dest: /etc/bind/named.conf
  fatal: [192.168.5.4]: FAILED! =>

- name: Настройка DHCP-сервера
  template:
    src: dhcpd.conf.j2
    dest: /etc/dhcp/dhcpd.conf
  hosts: servers

- name: Настройка веб-сервера Apache
  template:
    src: apache_config.j2
    dest: /etc/apache2/sites-available/000-default.conf
  notify:
    - restart apache
  package:
    name:

- name: Настройка межсетевого экрана (iptables)
  iptables:
    chain: INPUT
    source: 0.0.0.0/0
    destination: 192.168.5.4
    protocol: tcp
    destination_port: 80
    jump: ACCEPT
  state: present

handlers:
- name: restart samba
  service:
    name: smbd
    state: restarted

- name: restart apache
  service:
    name: apache2
    state: restarted

- name: Создание домена XSOFТ.LOCAL
  command: samba-tool domain provision --use-rfc2307 --realm=XSOFТ.LOCAL --domain=XSOFТ --adminpass=Secret123!

- name: Включение и запуск сервисов
  service:
    name: "{{ item }}"
    state: started
    enabled: yes
  loop:
    - smbd
    - nmbd
    - winbind
    - named
    - isc-dhcp-server
    - nginx

- name: Настройка firewall (разрешить Samba, DNS, DHCP, HTTP)
  iptables:
    chain: INPUT
    protocol: "{{ item.protocol }}"
    destination_port: "{{ item.port }}"
    jump: ACCEPT
  loop:
    - { protocol: tcp, port: 53 } # DNS
    - { protocol: udp, port: 53 } # DNS
    - { protocol: udp, port: 67 } # DHCP
    - { protocol: tcp, port: 80 } # HTTP
    - { protocol: tcp, port: 389 } # LDAP
    - { protocol: tcp, port: 445 } # SMB

```

```

Job for isc-dhcp-server.service failed because the control process exited with error code.
See "systemctl status isc-dhcp-server.service" and "journalctl -xe" for details.
areg@serverareg:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - LSB: DHCP server
   Loaded: loaded (/etc/init.d/isc-dhcp-server; generated; vendor preset: enabled)
   Active: failed (Result: exit-code) since Wed 2025-04-02 17:27:55 MSK; 7s ago
     Docs: man:systemd-sysv-generator(8)
    Process: 4329 ExecStart=/etc/init.d/isc-dhcp-server start (code=exited, status=1/FAILURE)

анр 02 17:27:53 serverareg dhcpd[4343]: before submitting a bug. These pages explain the proper
анр 02 17:27:53 serverareg dhcpd[4343]: process and the information we find helpful for debugging..
анр 02 17:27:53 serverareg dhcpd[4343]:
анр 02 17:27:53 serverareg dhcpd[4343]: exiting.
анр 02 17:27:55 serverareg isc-dhcp-server[4329]: Starting ISC DHCPv4 server: dhcpdcheck syslog for diagnostic
анр 02 17:27:55 serverareg isc-dhcp-server[4329]: failed!
анр 02 17:27:55 serverareg systemd[1]: isc-dhcp-server.service: Control process exited, code=exited status=1
анр 02 17:27:55 serverareg systemd[1]: Failed to start LSB: DHCP server.
анр 02 17:27:55 serverareg systemd[1]: isc-dhcp-server.service: Unit entered failed state.
анр 02 17:27:55 serverareg systemd[1]: isc-dhcp-server.service: Failed with result 'exit-code'.
areg@serverareg:~$ ^C
areg@serverareg:~$ cat /var/log/syslog | grep dhcpd
cat: /var/log/syslog: Отказано в доступе
areg@serverareg:~$ sudo cat /var/log/syslog | grep dhcpd
Двоичный файл (стандартный ввод) совпадает
areg@serverareg:~$

```

Возникла проблема с запуском Dhcp сервера после просмотра логов структуры сервера и где выдавалась ошибка принял решение поменять серив т к думаю проблема была в совместимости с астра линуксом

Удалил dhcp сервер из плейбука буду создавать отдельный

```
06:58:33 | 4 Apr, Friday | in → xsoft_infra
ansible-playbook -i inventory.ini 01_server.yml

PLAY [Настройка сервера и сервисов] *****

TASK [Gathering Facts] *****
ok: [192.168.5.4]

TASK [Проверка интерфейса] *****
ok: [192.168.5.4]

TASK [Вывод если интерфейс недоступен] *****
skipping: [192.168.5.4]

TASK [Установка пакетов] *****
ok: [192.168.5.4]

TASK [Настройка Samba] *****
ok: [192.168.5.4]

TASK [Создание домена] *****
ok: [192.168.5.4]

TASK [Настройка Bind9] *****
ok: [192.168.5.4]

TASK [Настройка Apache] *****
ok: [192.168.5.4]

TASK [Настройка iptables] *****
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 53})
ok: [192.168.5.4] => (item={'protocol': 'udp', 'port': 53})
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 80})
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 389})
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 445})

TASK [Запуск сервисов] *****
ok: [192.168.5.4] => (item=smbd)
ok: [192.168.5.4] => (item=nmbd)
ok: [192.168.5.4] => (item=bind9)
ok: [192.168.5.4] => (item=apache2)

PLAY RECAP *****
192.168.5.4 : ok=9 changed=0 unreachable=0 failed=0 skipped=1 rescued=0 ignored=0

on bash 0ms
```

После проверок решил удалить старый isc-dhcp-server и очистить порты после чего новый Dhcр сервис Dhcрmasq стал работать так же избавился от ненужного dns сервера bind9 потому что Dhcрmasq уже в себя включает и dhcp и dns сервера

Удаление isc-dhcp-server

```
areg@serverareg:~$ sudo ss -tulnp | grep -E ':53|:67'
udp      UNCONN      0      0      *:5353          *:*              users:((("avahi-daemon",pid=550,fd=12))
udp      UNCONN      0      0      192.168.5.4:53  *:*              user
s:((("named",pid=642,fd=520),("named",pid=642,fd=519),("named",pid=642,fd=518))
udp      UNCONN      0      0      127.0.0.1:53    *:*              users:
((("named",pid=642,fd=517),("named",pid=642,fd=516),("named",pid=642,fd=515))
udp      UNCONN      0      0      :::5353         :::*              users:((("avahi-daemon",pid=550,fd=13))
udp      UNCONN      0      0      :::53           :::*              users:((("named",pid=642,fd=514),("named",pid=642,fd=513),("named",pid=642,fd=512))
tcp      LISTEN        0      10     192.168.5.4:53  *:*              user
s:((("named",pid=642,fd=25))
tcp      LISTEN        0      10     127.0.0.1:53    *:*              users:
((("named",pid=642,fd=22))
tcp      LISTEN        0      10     :::53           :::*              users:((("named",pid=642,fd=21))
t -i:67) 2>/dev/null || echo "No processes on 67"areg@serverareg:~$
areg@serverareg:~$ # Убьём все процессы, занимающие порты (если есть)
```

Очистка портов

```
areg@serverareg:~$ systemctl -l | grep "dhcp"^C
areg@serverareg:~$ sudo systemctl stop isc-dhcp-server
areg@serverareg:~$ sudo systemctl disable isc-dhcp-server
cp/isc-dhcp-server.service is not a native service, redirecting to systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install disable isc-dhcp-server
areg@serverareg:~$ sudo apt purge isc-dhcp-server -y
Чтение списков пакетов... Готово
Построение дерева зависимостей
Чтение информации о состоянии... Готово
Следующие пакеты устанавливались автоматически и больше не требуются:
  isc-dhcp-common libirs-export160 libiscfg-export160 policycoreutils selinux-utils
Для их удаления используйте «sudo apt autoremove».
Пакеты, которые будут УДАЛЕНЫ:
  isc-dhcp-server*
обновлено 0, установлено 0 новых пакетов, для удаления отмечено 1 пакетов, и 0 пакетов не об-
новлено.
После данной операции, объём занятого дискового пространства уменьшится на 1 424 kB.
(Чтение базы данных ... на данный момент установлено 175803 файла и каталога.)
Удаляется isc-dhcp-server (4.3.5-3+deb9u1+ci202106011517+astra4) ...
Обрабатываются триггеры для man-db (2.7.6.1-2) ...
(Чтение базы данных ... на данный момент установлено 175785 файлов и каталогов.)
Вычищаются файлы настройки пакета isc-dhcp-server (4.3.5-3+deb9u1+ci202106011517+astra4) ...
dpkg: предупреждение: при удалении isc-dhcp-server каталог «/var/lib/dhcp» не пуст, поэтому
не удалён
```

```
Δ on bash
07:18:55 | 4 Apr, Friday | in → xsoft_infra
→ ansible-playbook -i inventory.ini 01_server.yml

PLAY [Настройка сервера и сервисов] *****

TASK [Gathering Facts] *****
ok: [192.168.5.4]

TASK [Обновление списка пакетов] *****
ok: [192.168.5.4]

TASK [Установка необходимых пакетов] *****
ok: [192.168.5.4]

TASK [Остановка systemd-resolved] *****
ok: [192.168.5.4]

TASK [Настройка Samba] *****
ok: [192.168.5.4]

TASK [Создание домена] *****
ok: [192.168.5.4]

TASK [Создание резервной копии конфига dnsmasq] *****
ok: [192.168.5.4]

TASK [Настройка конфигурации Dnsmasq] *****
ok: [192.168.5.4]

TASK [Создание лог-файла Dnsmasq] *****
changed: [192.168.5.4]

TASK [Настройка Apache] *****
ok: [192.168.5.4]

TASK [Настройка правил iptables] *****
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 53})
ok: [192.168.5.4] => (item={'protocol': 'udp', 'port': 53})
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 80})
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 389})
ok: [192.168.5.4] => (item={'protocol': 'tcp', 'port': 445})

TASK [Запуск и включение сервисов] *****
ok: [192.168.5.4] => (item=smbd)
ok: [192.168.5.4] => (item=nmbd)
ok: [192.168.5.4] => (item=dnsmasq)
ok: [192.168.5.4] => (item=apache2)

Onit dnsmasq.service could not be found.
areg@serverareg:~$ systemctl status dnsmasq.service
● dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server
   Loaded: loaded (/lib/systemd/system/dnsmasq.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2025-04-04 06:18:52 MSK; 4min 52s ago
     Process: 9236 ExecStartPost=/etc/init.d/dnsmasq systemd-start-resolvconf (code=exited, sta
     Process: 9225 ExecStart=/etc/init.d/dnsmasq systemd-exec (code=exited, status=0/SUCCESS)
     Process: 9222 ExecStartPre=/usr/sbin/dnsmasq --test (code=exited, status=0/SUCCESS)
    Main PID: 9235 (dnsmasq)
       Tasks: 1 (limit: 4915)
     CGroup: /system.slice/dnsmasq.service
             └─9235 /usr/sbin/dnsmasq -x /run/dnsmasq/dnsmasq.pid -u dnsmasq -7 /etc/dnsmasq.d

lines 1-10/10 (END)
```

Так выглядит плейбук для сервера на данный момент

Поменял Apache на Nginx

```
16:08:20 | 10 Apr, Thursday | in → xsoft_infra
ansible-playbook -i inventory.ini 01_server.yml

PLAY [Настройка сервера и сервисов] *****

TASK [Gathering Facts] *****
ok: [192.168.5.9]

TASK [Обновление списка пакетов] *****
ok: [192.168.5.9]

TASK [Установка необходимых пакетов] *****
ok: [192.168.5.9]

TASK [Остановка systemd-resolved] *****
ok: [192.168.5.9]

TASK [Настройка Samba] *****
ok: [192.168.5.9]

TASK [Создание домена] *****
ok: [192.168.5.9]

TASK [Создание резервной копии конфига dnsmasq] *****
ok: [192.168.5.9]

TASK [Настройка конфигурации Dnsmasq] *****
ok: [192.168.5.9]

TASK [Создание лог-файла Dnsmasq] *****
changed: [192.168.5.9]

TASK [Создание директории сайта] *****
ok: [192.168.5.9]

TASK [Настройка конфигурации Nginx] *****
ok: [192.168.5.9]

TASK [Активация сайта Nginx] *****
ok: [192.168.5.9]

TASK [Удаление дефолтного сайта Nginx] *****
ok: [192.168.5.9]

TASK [Настройка правил iptables] *****
ok: [192.168.5.9] => (item={'protocol': 'tcp', 'port': 53})
ok: [192.168.5.9] => (item={'protocol': 'udp', 'port': 53})
ok: [192.168.5.9] => (item={'protocol': 'tcp', 'port': 80})
ok: [192.168.5.9] => (item={'protocol': 'tcp', 'port': 443})
ok: [192.168.5.9] => (item={'protocol': 'tcp', 'port': 389})
ok: [192.168.5.9] => (item={'protocol': 'tcp', 'port': 445})
```

Начал писать плейбук для клиентов

```
11:20:59 | 12 Apr, Saturday | in → xsoft_infra
> ansible-playbook -i inventory.ini 02_clients.yml

PLAY [Настройка клиентов для подключения к домену] *****

TASK [Gathering Facts] *****
ok: [192.168.5.6]
ok: [192.168.5.5]

TASK [Установка необходимых пакетов на клиенте] *****
fatal: [192.168.5.6]: FAILED! => {"cache_update_time": 1743595799, "cache_updated": false, "changed": fa
et -y -o \"Dpkg::Options::--force-confdef\" -o \"Dpkg::Options::--force-confold\" install 'samba
.ce4' failed: E: Sub-process /usr/bin/dpkg returned an error code (1)\n", "rc": 100, "stderr": "E: Sub-
ed an error code (1)\n", "stderr_lines": ["E: Sub-process /usr/bin/dpkg returned an error code (1)", "s
ts...\nBuilding dependency tree...\nReading state information...\nsamba is already the newest version (2
e4).\n0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.\n4 not fully installed or removed.\
f additional disk space will be used.\nSetting up python-dnspython (1.15.0-1+deb9u1) ...\r\nTraceback (m
File \"usr/bin/pycompile\", line 35, in <module>\r\n    from debpython.version import SUPPORTED, debso
usr/share/python/debpython/version.py\", line 24, in <module>\r\n    from ConfigParser import SafeConfig
r: No module named 'ConfigParser'\r\nDpkg: error processing package python-dnspython (--configure):\r\n
stallation script returned error exit status 1\r\nSetting up glusterfs-common (3.8.8-1+deb9u1) ..\r\nTr
st):\r\n File \"usr/bin/pycompile\", line 35, in <module>\r\n    from debpython.version import SUPPORT
File \"usr/share/python/debpython/version.py\", line 24, in <module>\r\n    from ConfigParser import
tFoundError: No module named 'ConfigParser'\r\nDpkg: error processing package glusterfs-common (--config
ed post-installation script returned error exit status 1\r\nSetting up python-configparser (3.5.0-1) ...
all last):\r\n File \"usr/bin/pycompile\", line 35, in <module>\r\n    from debpython.version import S
\\r\n File \"usr/share/python/debpython/version.py\", line 24, in <module>\r\n    from ConfigParser i
oduleNotFoundError: No module named 'ConfigParser'\r\nDpkg: error processing package python-configparser
s installed post-installation script returned error exit status 1\r\nDpkg: dependency problems prevent c
amba depends on python-dnspython; however:\r\n Package python-dnspython is not configured yet.\r\n\r\nD
e samba (--configure):\r\n dependency problems - leaving unconfigured\r\nProcessing triggers for libc-bi
1733+astra4) ...\r\nErrors were encountered while processing:\r\n python-dnspython\r\n glusterfs-common\r
samba\r\n\" \"stdout_lines\": [\"Reading package lists\", \"Building dependency tree\", \"Reading state i
```

исправил проблему которая заключалась в конфиг парсере конкретно астра линукс

```
11:21:05 | 12 Apr, Saturday | in → xsoft_infra
> ansible-playbook -i inventory.ini 02_clients.yml

PLAY [Подключение клиента к домену] *****

TASK [Gathering Facts] *****
ok: [192.168.5.6]
ok: [192.168.5.5]

TASK [Обновление списка пакетов] *****
ok: [192.168.5.6]
ok: [192.168.5.5]

TASK [Установка пакетов для подключения к домену] *****
ok: [192.168.5.6]
ok: [192.168.5.5]

TASK [Настройка Kerberos] *****
changed: [192.168.5.6]
changed: [192.168.5.5]

TASK [Присоединение клиента к домену] *****
fatal: [192.168.5.5]: FAILED! => {"changed": true, "cmd": ["realm", "join", "--user=Administrator", "--password=Secret123!", "xsoft
local"], "delta": "0:00:00.382556", "end": "2025-04-12 10:21:19.354004", "msg": "non-zero return code", "rc": 2, "start": "2025-04-
2 10:21:18.971448", "stderr": "realm: Неизвестный параметр --password=Secret123!", "stderr_lines": ["realm: Неизвестный параметр --
password=Secret123!"], "stdout": "", "stdout_lines": []}
...ignoring
fatal: [192.168.5.6]: FAILED! => {"changed": true, "cmd": ["realm", "join", "--user=Administrator", "--password=Secret123!", "xsoft
local"], "delta": "0:00:00.555897", "end": "2025-04-12 10:21:18.998370", "msg": "non-zero return code", "rc": 2, "start": "2025-04-
2 10:21:18.442473", "stderr": "realm: Неизвестный параметр --password=Secret123!", "stderr_lines": ["realm: Неизвестный параметр --
password=Secret123!"], "stdout": "", "stdout_lines": []}
...ignoring

TASK [Проверка статуса подключения] *****
[WARNING]: conditional statements should not include Jinja2 templating delimiters such as {{ }} or {% %}. Found: '{{ domain }}' not
in realm_status.stdout
fatal: [192.168.5.5]: FAILED! => {"changed": true, "cmd": ["realm", "list"], "delta": "0:00:00.016403", "end": "2025-04-12 10:21:19
823063", "failed_when_result": true, "msg": "", "rc": 0, "start": "2025-04-12 10:21:19.806660", "stderr": "", "stderr_lines": [], "
tdout": "", "stdout_lines": []}
fatal: [192.168.5.6]: FAILED! => {"changed": true, "cmd": ["realm", "list"], "delta": "0:00:00.021970", "end": "2025-04-12 10:21:19
401257", "failed_when_result": true, "msg": "", "rc": 0, "start": "2025-04-12 10:21:19.379287", "stderr": "", "stderr_lines": [], "
tdout": "", "stdout_lines": []}
```

После множество ошибок и проблем с зависимостями астра линукс было принято решение доделать все на обычном ubuntu сервере скачал сsx подключился обновил пакеты установил пайтон для корректной работы с ансиблом теперь с заново напишу плейбук

```

reg@server:~$ sudo systemctl start ssh
sudo: systemctl: command not found
reg@server:~$ sudo systemctl start ssh
reg@server:~$ sudo systemctl status ssh
ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-04-13 09:57:16 UTC; 6s ago
     TriggeredBy: • ssh.socket
       Docs: man:sshd(8)
              man:sshd_config(5)
    Process: 2456 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 2457 (sshd)
      Tasks: 1 (limit: 7020)
     Memory: 1.2M (peak: 1.6M)
        CPU: 33ms
    CGroup: /system.slice/ssh.service
            └─2457 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Apr 13 09:57:16 server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Apr 13 09:57:16 server sshd[2457]: Server listening on :: port 22.
Apr 13 09:57:16 server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
reg@server:~$

```

```

PLAY [Настройка сервера Xsoft] *****
TASK [Gathering Facts] *****
ok: [192.168.5.13]

TASK [Обновление кеша и установка зависимостей] *****
ok: [192.168.5.13]

TASK [Остановить systemd-resolved] *****
ok: [192.168.5.13]

TASK [Удалить символическую ссылку /etc/resolv.conf] *****
changed: [192.168.5.13]

TASK [Установить локальный nameserver в resolv.conf] *****
changed: [192.168.5.13]

TASK [Удаление старого smb.conf перед инициализацией домена] *****
changed: [192.168.5.13]

TASK [Инициализация домена Samba (если ещё не выполнено)] *****
ok: [192.168.5.13]

TASK [Настройка DHCP] *****
ok: [192.168.5.13]

TASK [Настройка nginx] *****
ok: [192.168.5.13]

TASK [Применение конфигурации iptables] *****
ok: [192.168.5.13]

TASK [Обеспечение автозагрузки iptables правил] *****
ok: [192.168.5.13]

TASK [Перезапуск всех основных служб] *****
changed: [192.168.5.13] => (item=samba-ad-dc)
changed: [192.168.5.13] => (item=isc-dhcp-server)
changed: [192.168.5.13] => (item=nginx)

PLAY RECAP *****
192.168.5.13 : ok=12 changed=4 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

```

```
areg@srvareg:~$ sudo systemctl status isc-dhcp-server.service
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled;
   Active: active (running) since Wed 2025-04-16 15:27:22 UTC; 8min ago
     Docs: man:dhcpd(8).
    Main PID: 19267 (dhcpd)
      Tasks: 1 (limit: 7625)
     Memory: 3.7M (peak: 4.1M)
        CPU: 21ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─19267 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server
```

```
areg@srvareg:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: ena
   Active: active (running) since Wed 2025-04-16 15:27:22 UTC; 8min ago
     Docs: man:nginx(8).
    Main PID: 19304 (nginx)
      Tasks: 5 (limit: 7625)
     Memory: 3.7M (peak: 4.3M)
        CPU: 26ms
    CGroup: /system.slice/nginx.service
            └─19304 "nginx: master process /usr/sbin/nginx -q daemon on; master
```

```

areg@srvareg:~$ sudo systemctl status samba-ad-dc
● samba-ad-dc.service - Samba AD Daemon
   Loaded: loaded (/usr/lib/systemd/system/samba-ad-dc.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-16 15:34:08 UTC; 1s ago
     Docs: man:samba(8)
           man:samba(7)
           man:smb.conf(5)
   Process: 19967 ExecCondition=/usr/share/samba/is-configured samba (code=exited, status=0)
   Main PID: 19971 (samba)
    Status: "samba: ready to serve connections..."
   Tasks: 60 (limit: 7625)
  Memory: 228.2M (peak: 269.0M)
     CPU: 1.436s
   CGroup: /system.slice/samba-ad-dc.service
           └─19971 "samba: root process"

```

```

areg@srvareg:~$ echo "
dn=@,cn=_ldap._tcp,cn=serverareg.local,ou=DomainDnsZones,dc=serverareg,dc=local
changetype: add
objectClass: dnsRecord
objectClass: dnsSrvRecord
dnsRecord: 0 100 389 srvareg.serverareg.local
" | sudo ldbmodify -H /var/lib/samba/private/sam.ldb
Error: First line of ldif must be a dn not 'dn=@,cn=_ldap._tcp,cn=serverareg.local,ou=Domain
DnsZones,dc=serverareg,dc=local
changetype'
Failed to parse ldif
ldbmodify failed after processing 0 records

```

```

areg@srvareg:~$ sudo samba-tool dns query localhost serverareg.local _ldap._tcp SRV -U "admini
nistrator"
WARNING: Using passwords on command line is insecure. Installing the setproctitle python mod
ule will hide these from shortly after program start.
Password for [SERVERAREG\administrator]:
Name=, Records=1, Children=0
SRV: srvareg.serverareg.local. (389, 0, 100) (flags=f0, serial=1, ttl=900)

```

```

areg@srvareg:~$ host -t SRV _ldap._tcp.serverareg.local 127.0.0.1
Using domain server:
Name: 127.0.0.1
Address: 127.0.0.1#53
Aliases:

```

```

02:11:12 | 19 Apr, Saturday | IN | xsoit-initd
> ansible-playbook -i inventory.ini 02_clients.yml

AY [Настройка клиента и подключение к домену Samba AD] *****
SK [Gathering Facts] *****
: [192.168.5.15]
: [192.168.5.16]

SK [Установить необходимые пакеты] *****
: [192.168.5.16]
: [192.168.5.15]

SK [Конфигурировать Kerberos] *****
: [192.168.5.15]
: [192.168.5.16]

SK [Отключить AppArmor (если используется)] *****
: [192.168.5.15]
: [192.168.5.16]

SK [Настроить DNS – задать DNS-сервер вручную] *****
anged: [192.168.5.15]
anged: [192.168.5.16]

SK [Проверить SRV-запись контроллера домена] *****
tal: [192.168.5.15]: FAILED! => {"changed": false, "cmd": ["host",
serverareg.local"], "delta": "0:00:00.050998", "end": "2025-04-19
when_result": true, "msg": "non-zero return code", "rc": 1, "start
2583", "stderr": "", "stderr_lines": [], "stdout": ";; communicati
53: connection refused\n;; communications error to 192.168.5.14#53
o servers could be reached", "stdout_lines": [";; communications e
onnection refused", ";; communications error to 192.168.5.14#53: c
servers could be reached"]}

```

Начал редактировать плейбук для клиентов первые ошибки

Клиенты не видели сервер исправил

```

client1areg@client1admin:~$ ping 192.168.5.14
PING 192.168.5.14 (192.168.5.14) 56(84) bytes of data.
^C
--- 192.168.5.14 ping statistics ---
14 packets transmitted, 0 received, 100% packet loss, time 26059
client1areg@client1admin:~$ ping 192.168.5.14
PING 192.168.5.14 (192.168.5.14) 56(84) bytes of data.
64 bytes from 192.168.5.14: icmp_seq=1 ttl=64 time=0.574 ms
64 bytes from 192.168.5.14: icmp_seq=2 ttl=64 time=10.3 ms
64 bytes from 192.168.5.14: icmp_seq=3 ttl=64 time=1.55 ms
^C
--- 192.168.5.14 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 7588ms
rtt min/avg/max/mdev = 0.574/4.150/10.330/4.387 ms

```

Исправил ошибку в конф файле айпитейблс теперь успешно клиенты пингуются

Проверка подключения к домену.

```

client1areg@client1admin:~$ realm discover SERVERAREG.LOCAL
serverareg.local
  type: kerberos
  realm-name: SERVERAREG.LOCAL
  domain-name: serverareg.local
  configured: no
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin

```

Не удалось подключиться после просмотра логов решил проблему удалением из самбы клиентские машины которые я удалил и теперь использую другие но имя было у них одинаковое

```

client1areg@client1admin:~$ sudo realm join --user=administrator SERVERAREG.LOCAL
Password for administrator:
See: journalctl REALMD_OPERATION=r3891.5426
realm: Couldn't join realm: Failed to join the domain
client1areg@client1admin:~$ sudo journalctl REALMD_OPERATION=r3891.5426
Apr 19 18:40:25 client1admin realmd[4859]: * Resolving: _ldap._tcp.serverareg.local
Apr 19 18:40:25 client1admin realmd[4859]: * Performing LDAP DSE lookup on: 192.168.5.15
Apr 19 18:40:25 client1admin realmd[4859]: * Performing LDAP DSE lookup on: 2a00:
areg@srvareg:~$ sudo samba-tool ou create "OU=Computers,DC=serverareg,DC=local"
Added ou "OU=Computers,DC=serverareg,DC=local"

```

В плейбуке указал пароль без шифровки т.к. задачу ддля конкурса так бы я его зашифровал для безопасности

```

k: [192.168.5.15]
ASK [Показать результат SRV-запроса] *****
k: [192.168.5.16] => {
  "srv_check.stdout_lines": [
    "_ldap._tcp.serverareg.local has SRV record 0 100 389 srvareg.serverareg.local."
  ]
}
k: [192.168.5.15] => {
  "srv_check.stdout_lines": [
    "_ldap._tcp.serverareg.local has SRV record 0 100 389 srvareg.serverareg.local."
  ]
}
ASK [Выполнить kinit от имени администратора] *****
changed: [192.168.5.15]
changed: [192.168.5.16]
ASK [Присоединить клиента к домену с указанием OU] *****
changed: [192.168.5.15]
fatal: [192.168.5.16]: FAILED! => {"censored": "the output has been hidden due to the fact that 'no_log:'
  result", "changed": true}
ASK [Перезапустить службу sssd] *****
changed: [192.168.5.15]
ASK [Проверить подключение к домену] *****
k: [192.168.5.15]
ASK [Убедиться, что клиент успешно подключен к домену] *****
WARNING: conditional statements should not include jinja2 templating delimiters such as {{ stdout
  { }} or {% %}. Found: '{{ domain_name }}.local' in realm_status.stdout
k: [192.168.5.15] => {
  "realm_status.stdout": "serverareg.local\n type: kerberos\n realm-name: SERVERAREG.LOCAL\n domain
nfigured: kerberos-member\n server-software: active-directory\n client-software: sssd\n required-pac
package: sssd\n required-package: libnss-sss\n required-package: libpam-sss\n required-package: adcl
common-bin\n login-formats: %U@serverareg.local\n login-policy: allow-realm-logins"

```

Успешно подключился первый клиент

```
client2areg@client2:~$ realm list
serverareg.local
  type: kerberos
  realm-name: SERVERAREG.LOCAL
  domain-name: serverareg.local
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@serverareg.local
  login-policy: allow-realm-logins
client2areg@client2:~$
```

Первый клиент подключился со вторым возникли ошибки принялся исправлять

```

PLAY [Настройка клиента и подключение к домену Samba AD] *****

TASK [Gathering Facts] *****
ok: [192.168.5.15]
ok: [192.168.5.16]

TASK [Установить необходимые пакеты] *****
ok: [192.168.5.15]
ok: [192.168.5.16]

TASK [Конфигурировать Kerberos] *****
ok: [192.168.5.15]
changed: [192.168.5.16]

TASK [Отключить AppArmor (если используется)] *****
ok: [192.168.5.16]
ok: [192.168.5.15]

TASK [Настроить DNS – задать DNS-сервер вручную] *****
ok: [192.168.5.15]
ok: [192.168.5.16]

TASK [Проверить доступность домена] *****
ok: [192.168.5.15]
ok: [192.168.5.16]

TASK [Проверить текущий статус домена] *****
ok: [192.168.5.16]
ok: [192.168.5.15]

TASK [Выполнить kinit от имени администратора (если не в домене)] *****
skipping: [192.168.5.15]
changed: [192.168.5.16]

TASK [Присоединить клиента к домену (если еще не присоединен)] *****
skipping: [192.168.5.15]
changed: [192.168.5.16]

TASK [Перезапустить службу sssd (если были изменения)] *****
skipping: [192.168.5.15]
changed: [192.168.5.16]

TASK [Проверить финальный статус подключения] *****
ok: [192.168.5.15]
ok: [192.168.5.16]

TASK [Показать результат подключения] *****
ok: [192.168.5.16] => {
  "msg": "Итоговый статус: Успешно подключен к домену\n\nПодробности: {'changed': True, 'stdout': 'Password for administrator: ', 'stderr': '', 'rc': 0, 'cmd': ['realm', 'join', '--user=administrator', '--computer-ou=OU=Computers,DC=serverareg,DC=local', 'serverareg.local'], 'start': '2025-04-19 20:04:16.761405', 'end': '2025-04-19 20:04:21.622817', 'delta': '0:00:04.861412', 'msg': '', 'stdout_lines': ['Password for administrator: '], 'stderr_lines': [], 'failed': False}"
}
ok: [192.168.5.15] => {
  "msg": "Итоговый статус: Успешно подключен к домену\n\nПодробности: {'changed': False, 'skipped': True, 'skip_reason': 'Conditional result was False', 'false_condition': '\domain_status is failed or domain_name+'.local' not in domain_status.stdout}"
}

PLAY RECAP *****
192.168.5.15 : ok=9   changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.5.16 : ok=12  changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

Ошибка заключалась в кеше прошлых настроек КДС после очистки оба клиента успешно подключились к домену!

```
client1areg@client1admin:~$ realm list
serverareg.local
  type: kerberos
  realm-name: SERVERAREG.LOCAL
  domain-name: serverareg.local
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@serverareg.local
  login-policy: allow-realm-logins
client1areg@client1admin:~$
```

```
client2areg@client2:~$ realm list
serverareg.local
  type: kerberos
  realm-name: SERVERAREG.LOCAL
  domain-name: serverareg.local
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@serverareg.local
  login-policy: allow-realm-logins
client2areg@client2:~$
```

показывает успешную авторизацию Kerberos получил билет.

```
client1areg@client1admin:~$ id administrator@serverareg.local
uid=1133200500(administrator@serverareg.local) gid=1133200513 groups=1133200513
client1areg@client1admin:~$ klist
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: administrator@SERVERAREG.LOCAL

Valid starting    Expires          Service principal
04/19/2025 19:51:39  04/20/2025 05:51:39  krbtgt/SERVERAREG.LOCAL@SERVERAREG.LOCAL
renew until 04/20/2025 19:51:37
```

. Веб-сервер nginx отвечает.

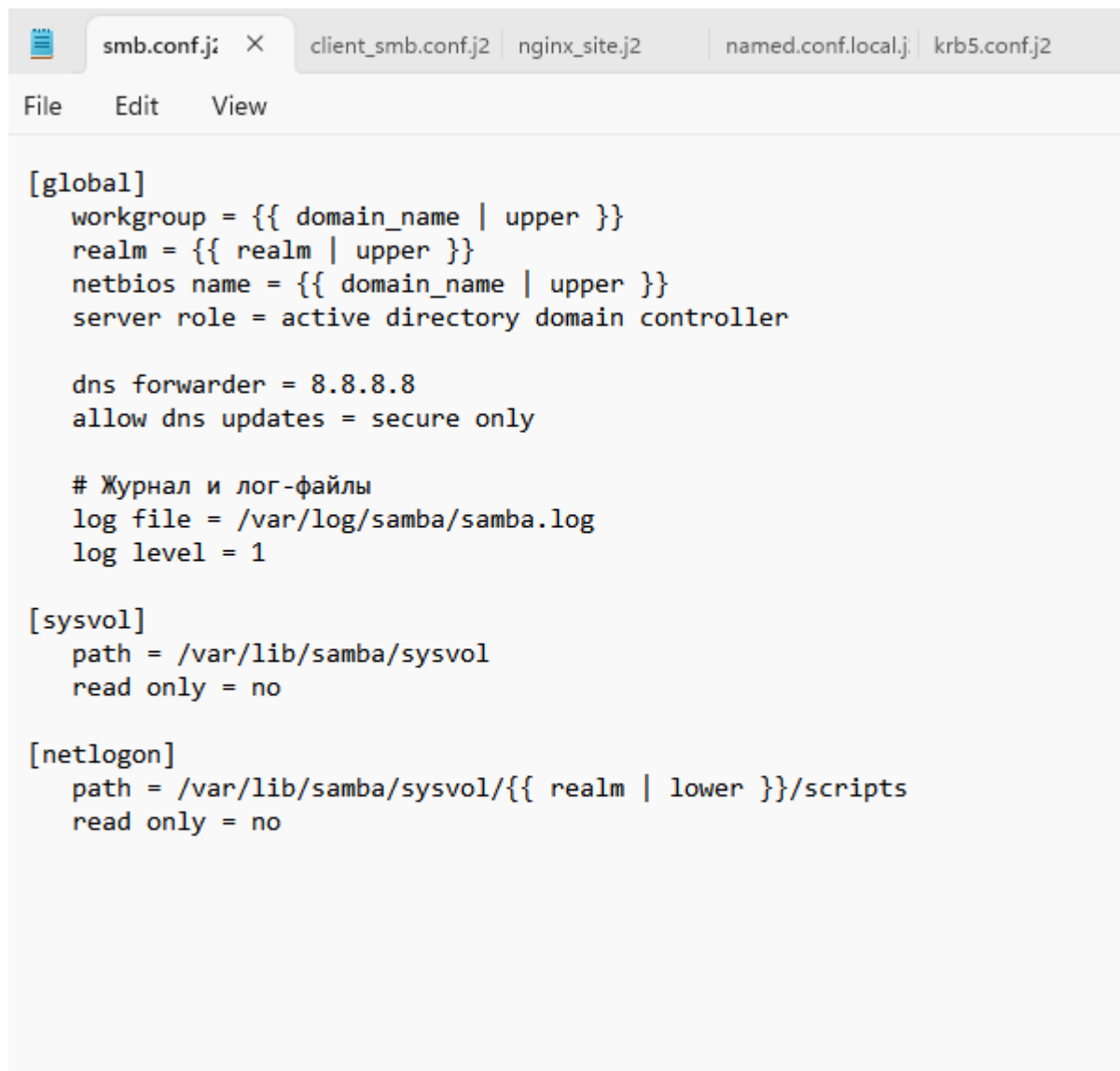
```
client1areg@client1admin:~$ curl http://192.168.5.14
<html>
<head><title>403 Forbidden</title></head>
<body>
<center><h1>403 Forbidden</h1></center>
<hr><center>nginx/1.24.0 (Ubuntu)</center>
</body>
</html>
client1areg@client1admin:~$
```

Спасибо за интересную задачу справился как смог. С Отечественным ПО как видите на астра линуксе очень много сервисов конфликтовали с записимостями делал что мог в конце все довел до рабочего формата чтобы все технические части конкурса работали.

Геворгян Арег

Вот все файлы конфигураций которые я использовал?

Smb.conf

A screenshot of a text editor window showing the configuration file smb.conf.j2. The window has a title bar with a file icon and several tabs: smb.conf.j2 (active), client_smb.conf.j2, nginx_site.j2, named.conf.local.j, and krb5.conf.j2. Below the tabs is a menu bar with 'File', 'Edit', and 'View'. The main area contains the Samba configuration in a monospaced font. It starts with a [global] section containing settings for workgroup, realm, netbios name, server role, dns forwarder, and logging. This is followed by a [sysvol] section for the sysvol path and read-only status, and finally a [netlogon] section for the netlogon path and read-only status.

```
[global]
    workgroup = {{ domain_name | upper }}
    realm = {{ realm | upper }}
    netbios name = {{ domain_name | upper }}
    server role = active directory domain controller

    dns forwarder = 8.8.8.8
    allow dns updates = secure only

    # Журнал и лог-файлы
    log file = /var/log/samba/samba.log
    log level = 1

[sysvol]
    path = /var/lib/samba/sysvol
    read only = no

[netlogon]
    path = /var/lib/samba/sysvol/{{ realm | lower }}/scripts
    read only = no
```

A screenshot of a text editor window with multiple tabs. The active tab is 'client_smb.' and it contains Samba configuration settings. The settings are listed under the '[global]' section and include workgroup, security, realm, password encryption, idmap configuration, shell template, winbind settings, and client signing.

```
[global]
workgroup = {{ domain_name|upper }}
security = ADS
realm = {{ realm }}
encrypt passwords = yes
idmap config * : backend = tdb
idmap config * : range = 3000-7999
template shell = /bin/bash
winbind use default domain = yes
winbind offline logon = false
client signing = yes
```

```
libdefaults]
default_realm = {{ realm | upper }}
dns_lookup_realm = false
dns_lookup_kdc = true

realms]
{{ realm | upper }} = {
    kdc = {{ server_ip }}
    admin_server = {{ server_ip }}
}

domain_realm]
.{{ domain_name }} = serverareg
{{ domain_name }} = serverareg
```

```

*filter
filter

# DNS
-A INPUT -p tcp --dport 53 -j ACCEPT
-A INPUT -p udp --dport 53 -j ACCEPT

# Kerberos
-A INPUT -p tcp --dport 88 -j ACCEPT
-A INPUT -p udp --dport 88 -j ACCEPT

# LDAP
-A INPUT -p tcp --dport 389 -j ACCEPT
-A INPUT -p udp --dport 389 -j ACCEPT

# SMB / NetBIOS
-A INPUT -p tcp --dport 139 -j ACCEPT
-A INPUT -p tcp --dport 445 -j ACCEPT

# RPC services
-A INPUT -p tcp --dport 135 -j ACCEPT

# Разрешаем трафик на локальном интерфейсе
-A INPUT -i lo -j ACCEPT

# Разрешаем трафик на порты для ssh и http
-A INPUT -p tcp --dport 22 -j ACCEPT
-A INPUT -p tcp --dport 80 -j ACCEPT

# Разрешаем ICMP
-A INPUT -p icmp --icmp-type 8 -j ACCEPT

# Разрешаем установленные соединения
-A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

# Блокируем все остальные входящие соединения
-A INPUT -j DROP

COMMIT

```

```

# Файл конфигурации LDAP клиента

# Адрес LDAP сервера
URI ldap://{ ldap_server }}

# Базовый DN (Distinguished Name)
BASE {{ ldap_base }}

# Режим поиска
TIMELIMIT 10
SIZELIMIT 500

# Параметры для подключения
LDAPVERSION 3
TLS_CACERT /etc/ssl/certs/ca-certificates.crt

# Указываем DN для подключения (если нужно)
# bind DN (если требуется аутентификация)
BINDDN {{ ldap_binddn }}

# Пароль для bind DN
BINDPW {{ ldap_bindpw }}

# Устанавливаем включение/выключение TLS
# Включаем TLS (если требуется)
TLS_REQCERT allow

```

```
$TTL      604800
@         IN      SOA      ns1.areg.local. admin.areg.local. (
                                1          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL

@         IN      NS       ns1.areg.local.
12        IN      PTR      server.areg.local.
```

```
$TTL      604800
@         IN      SOA      ns.{{ domain_name }}. admin.{{ domain_name }}. (
                                2          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ns.{{ domain_name }}.
ns        IN      A        192.168.5.12
```

File Edit View

```
server {  
    listen 80;  
    server_name {{ domain_name }} {{ server_ip }};  
  
    location / {  
        root /var/www/html;  
        index index.html index.htm;  
    }  
  
    # логи  
    access_log /var/log/nginx/access.log;  
    error_log /var/log/nginx/error.log;  
}
```

File Edit View

```
zone "{{ domain_name }}" IN {  
    type master;  
    file "/etc/bind/db.{{ domain_name }}";  
};  
  
zone "{{ realm | upper }}" IN {  
    type master;  
    file "/etc/bind/db.{{ realm | upper }}";  
};
```